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Content

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Description

2016 was the year of Java deserialization apocalypse. Although Java Deserialization attacks were known for years, the publication of the Apache Commons Collection Remote Code Execution gadget (RCE from now on) finally brought this forgotten vulnerability to the spotlight and motivated the community to start finding and fixing these issues. .NET is next in line; formatters such as BinaryFormatter and NetDataContractSerializer are known to share similar mechanics which make them potentially vulnerable to similar RCE attacks. However, as we saw with Java before, the lack of RCE gadgets led some software vendors to not take this issue seriously. In this talk, we will analyze .NET serializers including third party JSON parsers for potential RCE vectors. We will provide real-world examples of vulnerable code and more importantly, we will review how these vulnerabilities were detected and fixed in each case.

Transcript

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thank you can you hear me yeah so alright thank you Jim for the introduction so as Jim said I'm algorithm unit also known as pond tested in social network Twitter github and so on I work as a principal security researcher with fortify team I mainly do static analysis related research and probably well I focus on on web application security right why find more interesting is for getting remote code execution on web applications and I have responsibly reported critical remote code execution boon abilities to companies like Microsoft Oracle workday Salesforce and so on so one of them in Microsoft I will be sharing with you today hopefully if we have time so apart from that I have some experience around digitalization I started not in 2015 when as Jim said but actually in 2012 playing with XML decoder and access stream trying to get remote code execution in restful Web Services right so we were able to get remote code execution and some libraries like Restylane and rustici and then I moved into the binary format of Java visualization right and I found too worried to my knowledge is the first remote code execution gadget for Java deserialization but for the binary format back in 2013 and it was not in a patchy commerce collection but actually in the spring libraries right so unfortunately this gadget didn't become as popular as the Apache Commons collection gadget became and I'm saying unfortunately because we all saw that this was like a tipping point right and when Chris for hope and Gabriel Lawrence family's cut yet the community started looking for these vulnerabilities they started fixing them and so on so we could have saved ourselves like a couple of years by starting back in 2013 or even even before so at this point in 2015 Chris Roth and Gabrielle Oryza published the Apache Commons collection gadget and everything changes right all the researchers start looking into Java this realization they start publishing vulnerabilities like in WebLogic websphere and many other applications so I retook my interest in Java serialization I I found like a technique to bypass what is known as the Luca head object digitization that is like the standard mitigation technique for this kind of attacks in Java and also I found like multiple gadgets to lead into remote code execution including one in the Java Runtime itself so not requiring any third-party libraries in order to work so then I switch gears and star appeared looking into JSON to close the circle right so from XML to binary and then to Jason and I publish some approaches to get remote execution in JSON deserialize errs back in last year in black hat on and DEF CON so most of this is Java right and this is what is known as the Java DC allocation apocalypse and I think well it's bad because I mean there was many application that were affected by these bugs but in the other hand it was good because we got this categorical together to go into the OWASP top 10 right and we as a security community we have knowledge about the latest trends and latest categories like I don't know SPL injection or jndi injection but developers don't normally have these visibility in what boon abilities they should focus on and I think that having secured this realization in the August of 10 is a good thing in order for developers to start looking and fixing these vulnerabilities so at that point I decided to take a look into that net because well I'm not about net developer and not a dotnet researcher but since I had some previous experience with this realization I decided to take a look because apparently there

was a previous research around .NET visualization but there was no remote or execution gadget and has happened with Java before it was not until a remote code execution gadget was published that the community started fixing and taking care of this vulnerabilities so I decided to look for one in order to start you know the ball moving so what I'm going to present today is the result of these research so I will first introduce brief introduction into a serialization then we will be reviewing what are the most common doesn't .NET serialize have some formatters and then we will move to the more like builder and defender part of the top that is finding vulnerabilities and fixing them alright so Marceline pickles I will be using Rick and Morty picture because why not so this is Rick right he's an instance an object living in the VM and at some point the developer want to take Rick out of the VM right so he wants to put it in the file system or into the database or send it across the network so he put a rig across the magic portal and then he gets a pickle ring so at a later stage we want to recover the object back in the VM so we will put pickle Rick back into the magic portal and then recover and reconstruct Rick again alright so there is one thing in this slide that is critical for this DC allocation attacks to work anyone something that is like the design flow of all these formats like death so say that again sorry yeah well this is just a very simplification a simplified model of visualization but the issue I was referring to is that we're not sending just the pickle itself but we are sending a pickle week as long I love with the satellite data as well so this is what is known as the type discriminator and basically if you're familiar with Java additional ization even though it's a binary format you will find like class names or type names in the industry lies data same thing with dotnet for example this is JSON Dannette you will find some dollar attributes there are type attributes containing something that looks like dotnet type and dotnet assembly names so if the attacker can control the type discriminator and they change picker rig with a pickle morty then the server will blindly trust that and then will instantiate an instance of Mort instead of the of the brick one right so there is probably nothing really dangerous here because they're going to invoke the default constructor of this morty class which is it takes no parameters so there is nothing where the attacker can modify the behavior of this and just get arbitrary code execution however apart from the default constructor there will be other methods will get invoke like for example in Java will get the read object and with result methods and in .NET the same thing will have like this initialization constructor over here visualization callbacks even some annotations that we can use to annotate methods that are going to be involved with the framework in order to fully reconstruct the object right same thing with XML and JSON the this libraries will normally just invoke the default constructor and then invoke setters in order to populate the data in the class types and we they will be invoking centers in order to do that all right so all that the attackers need to do is find a gadget right and a gadget is nothing else that a type or a class that is sitting in the target application class path so they provide implementation for the precise they can only provide the type name and since they can control the values for these properties of this class then they are able to actually modify the behavior of the methods that are going to involve to be invoke like for example the deserialization callbacks or the setters so this is more like a builder and defender talk and I'm not going to focus on how to find and get final gadgets we live in two remote code execution but I think it's good to whom show you an example of a very simple one live in two remote code execution so you can have like an

idea of how they look like so this is a system windows data object data provider is app type in the windows guard so is available without any third-party dependencies required and it contains three different setters right that will initiate some invocations that will end up invoking this method over here and here we are basically just using reflection to invoke method name on an object instance passing some parameters right and those are exactly the one that we can control if we can control the values that are passed to the setters right so if an attacker provides a piece of JSON the net data like this one to a vulnerable endpoint the vulnerable endpoint will basically instantiate the object data provider and then it will invoke the setters for the object instance for the method parameters and for the method name and as we saw before this will initiate this this calls here and we'll end up invoking the reflection and getting the in this case the calculator to papad oh right so I'm not going to talk more about gadgets if you want to check which gadgets are available for Java obviously the white serial project if you want to know more about what gadgets internet I grow this project that is called why social net that basically it can be used to generate payloads and gadgets for different formats in the dotnet family so this is a simple example for example just you walk it like the one that we just saw if you want to attack the nation on that just specify the format there specify the Gardea to specify the command and it will generate the payload for you to use in your penetration test and pocs and so on so with that let's move into reviewing what format herbs and Syria lasers are vulnerable in they donate family so as I said this is nothing new and back in 2012 James Forster from the Google project zero was already presenting on how passing untrusted data to binary formatted or net data contracts dialyzers could live into some malicious code being run at that time there was no remote code execution gadget and I think that he only presented a Katia that was able to delete files from the file system or initiate some SM bid relay attacks right so has happened with Java before since there was since there were no remote code execution cut yet no one paid really you know attention to these vulnerabilities and there were like many many vulnerabilities are related with dotnet visualization until until last year probably so last year in 2017 my colleague Alexander meters and myself we found our remote code execution gadget in dotnet and that was followed by James for so presenting two more so there like a number of them right now so as we said James presented a couple of four matters that are vulnerable in their default configuration those are binary format there and net data contract serializer but they are more right these are the ones like I said that are vulnerable in their default configuration they should never be used with untrusted data at least in this default configuration and they include the binary formatter one that is the one that James presented but this one is also used internally by other serializers right for example this binary message format that is used by the Microsoft message queue system so if you can send them arbitrary types you will be able to get arbitrary code execution same thing with the object state formatter that is used by the asp.net view state so if you can change the view state and survive the h mark verification that will also lead to remote code execution and we will more about this later so then we have more like sub format that is using web services for JSON sweet Jason those are like third-party libraries so the most popular and used one is the binary formatter and this is probably the most common and P pattern that I found I literally found like thousands of applications in southern vulnerable to remote code execution because they were using

exactly this pattern and this is basically well this is very simple they are basically instance here in the binary format there and then say they're licensed some data basics de for encoding that data and putting that into a cookie so if that cookie came back the server is going to deserialize that and since there is no H Mac verification it will go in and run arbitrary commands using these gadgets so the first time that I found that was in app Harbor and one nose up Harbor yeah I know you use it so well browse on the side I realized that they were setting this cookie with this AE AE d that is nothing else that the magic number for the binary format there and then basic ste for encoding so that means that this side that is basically like Heroku for for donut applications like to host donut applications I was born about to remote code execution we reported that and they fix it but then we realized that they even had a blog post explaining how to do exactly what they were doing so instead of keeping some objects in the server side just saved some memory and put them in the client side by sterilizing them and putting them into a cookie so we tell them about this blog post and they are now deleted these this blog post however this same advice is everywhere you can find like multiple and yeasts and blog posts and even in Microsoft asp.net MVC framework was using the very same approach that they silently removed back in 2013 right and this is relatable with what Jeremy was talking before like if they don't publish like a CV German you don't know that there is something that you should update and so on this is kind of important so let's see a demo this is a vulnerability that I found in Asia yeah so this is from last year so as you can see even though this research by James first was published in 2012 there were many developers even Microsoft developers using binary format there with untrusted data so if you are familiar with Asia or Active Directory or application proxy you basically have your intranet here and you have the web application that you want to do sir right to to expose to your remote employees so your employees will basically connect to the application proxy and will be redirected to the AC or Active Directory well they will be like ask for the credentials that will be checked against your own instance of the active Active Directory and then they will be returning a token and then authenticating against the application proxy I think that is very better explained with a video so this is where visiting a application that I'm exposing through this is your application proxy and I'm ready ducted to the Microsoft single sign-on right I don't even need to login into a sure because this is a pre authentication remote code execution so I will basically send a request through a burp or any proxy and well basically I will just send this request to the repeater so we can see what is going on and as you can see here we get a 302 red erection to the single sign-on page in Microsoft and then we get this cookie assign here right if you recognize this magic number the same one in app Harbor this basically means remote code execution so we just get what we have to do now is generate our our payload and we will be basically this was before I wrote the white social donut so I had to do it by hand but basically I will be doing a DNS resolution against my own DNS server and I will be leaking the computer name just to prove that the remote code execution was not in my own intranet servers but actually in Asia servers so with that I will just generate the payload during the process of generation of generating the payload the payload will get executed so I will get a connection back to my DNS server from my own VM as you can see here so that's my VM computer name so nothing interesting so far but now if we copy paste the payload generated by our Cod yet or payload generator into

the cookie and we send this request to Microsoft Word to Lateral Network we will see that we will get some connection back from different servers actually not just one so for example the first one is from this server down here that is Microsoft servers and then if I submit the request again I will get a different connection which means that I can compromise all the application proxies that are working like in a load balancer way which means that I can intercept any traffic going to any intranet of customers using this a sure thing so that was pretty bad the good thing is that Microsoft only had to patch their own system and that was quite easy and they did it very quickly all right so apart from those DC realizers that are vulnerable in their default configuration there are other said allies errs that are known as template this a realizers that work with an expected type right so if you want to decide allies something they will only work with one type right for example if you're expecting a shopping cart instance then you cannot send like an object data provider payload because that won't work right so those examples examples of that is for example the data contract serializer that there are contract jesse realized and the xml said serialize so in these cases they are still vulnerable but the attacker needs to control not just the data that is going to be dis alized but also the suspected type you may think that this is not very common but I found out for example one good example in DotNetNuke that is the most popular content management system in in.net and basically at some point they were taking the contents of a cookie that was not sign it and that was like an XML cookie I don't know why but anyway they were extracting some type name from this XML and then they were invoking the constructor of the XML satellizer with these data that can be controlled by the attacker so now the attacker can control the expected type and then it will it was extracting the rest of the XML and then deserializing that so now that I can control the expected type and the data being DC realized which means that the attacker can control the whole object graph and run an arbitrary code in the server so that was reported and was fixed okay so the third group is those serializers that are vulnerable in some specific configuration right so those are for example JavaScript serializer that is the native microsoft don't JSON deserialize ER and json dot net that is the third party one that is even more popular than the Microsoft one so let's see the first one this is JavaScript serialize a javascript serialize it is safe to be used with untrusted data as long as you don't do this that is basically passing a type resolver if you use a type resolver that means that is going to include the type information along the with a JSON data and then is going to process that type information so it's going to allow the attackers to are to instantiate arbitrary types and invoke these setters that will lead to arbitrary code execution same thing with JSON dotnet but instead from an argument to the to the constructor in this case is a property right and they are set in this type name handling which means that it will include and process the type information in the JSON data so you have to set known because if you set objects arrays or router then the framework will be including this information and it will make the the endpoint vulnerable alright so how about finding if your applications are vulnerable so they are like different approaches there the first one probably the most simple one is to inspect the traffic flow in in and out of your applications and look for magic numbers or these signatures of the traffic being sterilized data as we saw before this a EAD is an example of this this is the basis basics t4 format for the magic number of the binary formatter but there are other magic numbers like for example for the object

state format error for the XML serialization and there are other encoding so I wrote this simple bear plug-in that you can freely use and basically we will use I will check for this magic magic numbers in different encodings so what I didn't meant to be like a perfect very accurate plug-in so it will have some false positives like for example if this magic number is found in in some images that will be reported and also it might find that you are sending said a lies data but it's not checking if there is a signature for that serialize data so maybe you cannot exploit it it's not a false positive because there is sterilized data but it's not straight about so we will see more about the viewstate later so the second approach is an active approach where you are basically sending a payload and watching if that payload gets executed and trying to recognize out-of-band if that gets executed so example of this is like for example making the server asleep for our number of seconds and then check if the response gets delayed or for example like I did with a sure you can run at the ns look up in order to if you control the DNS server to check if the payload is been executed IIST is another approach I don't know if it's passive active or something in the middle but I consider it active because you can you have to actually instrument your filter machines right so you instrument the virtual machines do you monitor what is going on inside of the virtual machine and then if some untrusted data gets into this deserialized method then we will report that that's good for pre-production but not for production anyway if you have access to the source code that's probably the easiest approach just check if some and think that there are some untrusted data flows into one of these three serialize things if you do that you have to consider that it's not as simple well for some for matters like binary format it is very simple first for some others for matters such as XML serialization you have to correlate different data flows and also taking a control flow into account because there can be a serialization binder that we will see later so you have to consider that but this is something that most of the static analysis tool can do so if you use fortify I know it worked because I brought the support for that so you are saying also fixing vulnerable endpoints right so now you find that you're the found that your application is vulnerable sorry so how can you fix it right so the first approach is do you really need to sterilize that data and this seems silly but I found like many application that were using serialization for Strings right and one good case for for this and one good example is the Nancy web application framework the Nancy is basically a web application framework that is very similar to Sinatra framework in in Ruby and they care about security and they implemented this hard protection but instead of putting a unique token into the safe cookie they were sterilizing that token into binary format sterilized data and then putting that into the cookie so making the whole framework and then all the applications build using this framework vulnerable to remote execution because of this implementation of the CSR protection same thing with type discriminators and JSON and XML is okay to to use but you should not use it we use them with type information right so that was the case of the breeze framework breeze is a dotnet entity management framework they basically suppose it's very similar to spring data rest so they expose some database entities through our restful web service automatically and they were sending the type information to the JavaScript client so that made no sense because javascript is not going to understand the dotnet type so they were not using that so because of these a mistake the whole framework was vulnerable to promote code execution and they fix it very quickly in a

matter of hours by just setting this type name handling setting to known right so this is actually the default value so probably at some point they were playing with that and they change that and they remain in the code so I think it's a good idea to explicitly stab these secure values explicitly and then even add a comment like do not change this please so okay well if you really need to serialize data so the first case was when do were satellites in a stream so that was not required now if you have some object graph that you want to persist in the client side for example for performance reasons or whatever reasons then if you're not expecting the client to modify that data then sign it I'm verified that the data hasn't been modified right if you do that always use an H Mac and never use something like md5 or sha-1 with secret and data because that's what about two has length extension attacks and this is the way that app Harbor and issuer actually fix their issues and they just sign the the data that they were not expecting anyone to change and then they verify that touch ends didn't get modified same thing with asp.net if you remember back in 2013 they removed this cookie temp data provider and they replace it with something else and that was basically the same approach of sterilizing server-side data into the client side but this time they were using the data protector API right they were doing something very similar to this call the data protector API protected by it and that will basically sign and encrypt the data using the web config validation and encryption keys and then this data won't be able to be tampered with without the server-side noticing so another good example of sign-in I'm verifying that the data is the asp.net viewstate right so for those of you that are not familiar with the viewstate it's like a representation of the state of a page so the server is going to send you the state of the page to you as a hidden field for the form field and then when you submit the form back you are submitting their previous state so the server can reconstruct a previous state and then continue from there so this is used by asp.net and I think that in the beginning they were not even forcing or they were not signing and encrypting the view state so their problem back in these days was that if there was some sensitive data for example in the viewstate then the clients were going to be able to extract and dig that data also if the client was able to modify the data because it was not it was not sign it and then they could change the application logic so that was back in the good days like many years ago and at some point in version 4.5.2 Microsoft even started ignoring the enable view State Mac meaning that even if you set that to false they will always encrypt and sign your view state which is good and they even back ported that back to 1.1 so I was curious and I run Arshavin query and I found that more than 200 servers is still running vulnerable version that are not using any signature or encryption in the view state which makes like a good cryptocurrency botnet or something like that anyway at some point they improve the cryptographic posture of the abuse state by actually using a key derivation function and this was a very interesting approach because if the attacker is able to leak the validation and encryption key then they will be able to actually change the encryption and change will not change the christian but changing the view state and then put in the payload there and get our query code execution so what they did is that they generate some per request are unique strings that they call purposes and what they do is that they run these through a key DF a key derivation function in order to generate some keys that are unique for each request now if the attacker is able to link this encryption and validation keys those kids won't be able to

tamper and modify the view state because the ones that will be used are these ones over here right so I just you know connected a debugger and took a look and I visit the account register page and then I check the values for these purposes strings so the first one is primary purpose right and this one is always going to be the same is this over here with the client state if he's for the viewstate other values for other things and then you have these specific purposes right so the first one is a template source directory a slash account the second one is an account register.aspx so both of them quite predictable for an attacker right the third one looks like pretty random generated like these seven seven six easy whatever value that looks like arbitrary value however this is the same value that is used in your anti cross-site request forgery token cookie right so the attacker is going to be able to read that and it's going to be able to generate all these strings and therefore if they are able to leak this case then they will be able to generate those keys right so now the problem is you have to be very careful with not leaking the case the first way that the attacker can actually lick the keys is if they find a file disclosure vulnerability in your application like for example and xxe vulnerability also they may be some vulnerabilities in the framework like in 2010 and 15 that allow the attackers to actually leak those keys from Troy's side you may recognize these apparently you can actually leak the validation key and decryption key if you are not using custom errors at least in the past also I found like hundreds of validation and decryption keys in github so if you are going to install something like a CMS or a block just by you know cloning the repo from from github and then installing that be very careful and always generate nucleus never use the ones that are provided by these kind of projects same thing with one-click installer do you get the latest version of these whatever CMS or whatever block application and then you stole it and some of them will use static keys that they won't generate per installation in those cases the attacker will be able to install the same at CMS for example for example and then ma'am you know no lick the keys and then attack and get arbitrary code execution in yours EMS block application or whatever so be very very careful with leaking the key a good advice is to encrypt some sections of your web config file so you can actually ask a is to encrypt this section so if you get like xxe vulnerability or a file displeasure vulnerability at least they won't be able to leak those keys so sign-in is okay if the client doesn't need to modify the data so that's that's that's okay but what if this is a pure like client server that are sending like arbitrary messages then you cannot sign the the data because in that case the client needs to have the signature keys the validation keys which if the attacker is in control of the client it makes no sense so these in this case what you have to do is actually bind it right this is very simple it's very similar to the concept of the look-ahead visualization in Java so basically right before instantiate in those types that are controlled by the attacker just stop for a while and ask yourself if you want to really be serialize yes this assembly name and this type name right so you are provided with this assembly name and type name and you can stop and apply a white list or a black list and then decide if you want to continue with the digitalization or stop right so this is a done by implementing the serialization binder interface this is a good example by jonathan built from Microsoft basically just implement this interface and then prepare like a list of allowed types with the type names that you are going to allow and then a pre-loaded instance of those types this will be important I will show you why later then just implement they override their bind to

type and then check if the type that the attacker is sending is what is within the white list of allowed types and if that's the case then continuum returned the preloaded type otherwise just throw an exception and aboard the the visualization so then you have to actually instantiate the serialization binder assign it to the binder property and deserialize right easy so while I was reviewing some Microsoft product I found one of them being vulnerable to remote code execution through visualization and they had something they have a additional ization binder in place so it was like okay they arcane about this but then I realized that the serialization ban there was not that secure as they thought for example so this is not the actual Microsoft code but this is something that I've put together in order to reproduce more or less what they were doing so I'm still in the process of this closing this one everything I haven't been fixed so at some point what they were doing is they were checking if the namespace starts with Microsoft dot put the name of a very popular Microsoft product here and if that's the case then load the type and return it otherwise return neuro so there are multiple vulnerabilities here first of all if you return null it may not do what you may think that is doing it's not going to avoid the desal ization it's going to fall back to the default binder which will bind anything which will bind including the remote code execution gadget type so it's not I am not protecting anything so the second problem here is that loading the type using reflection at this realization time is very costly and actually this will lead to the OS the attacker can easily do it your application because you are not using pre-loaded types if you remember the good example from from jonathan he was actually pre loading the types here and then just returning them in a method that gets executed you then did this realization and the third problem here is that this space this namespace here is just huge it contains like thousands of applicable of types right so even if the attacker cannot find a remote code execution gadget within this Microsoft dot something namespace they will be able probably chances are that they will be able to find what is known as a bypass and cut yet and what is a bypass guard yet it's just a gadget so it starts with a desire a station callback and at some point instead of running arbitrary code for example or doing bad things they just run nested this realization operation the problem here is that even though you are protecting your outer digitization operation the inner one the one that is run by this bypass cut yet has no binder in it right you have its instantiated here and then just this realizing there is no binder assignment so that means that this distillation operation will let all the remote code execution graduate types to be processed and deserialize and and the pillows to be executed so yeah this is basically some apps what I was talking about never return null because that this will fall back to the default binder and that's bad also don't use reflection to load types at runtime because that will do as your application and the fourth approach is to replace it right so we so the first one is stop using it if you don't need it the second one if the client doesn't need to change the data then sign it and verify that third approach is if the client needs to actually send you data then use a binder to control what types gets digitized and the fourth approach is if the architecture of your application allows you to do that change these formats that include the type discriminators the type information change that with something else and a good alternative here is Google protocol buffers where do you agree the client and the server agree in what they are going to exchange and then the exchange does the data so if the urban talks change an instance of a comb dog acme dot user then

they will agree they will know in advance that they are going to exchange that and then they will just share the name of the user the last name of the user the address of the user but no type information no property type names or much like that and also if you don't really need the type information like we said before just don't use it and that will make those JSON XML sterilizers to be safe at least from this kind of attacks so with that mahalo and if you have any questions I think we have time right to take questions [Music] yeah yeah yes that's that's this approach and and basically the only cases where you really need include type information that I can think of is if you are dealing with for example polymorphism types where you may send I don't know an instance of an animal but that may you need the information to need to to know if you need to reconstruct a dog or a cat for example so you need to send information the type information with that and that's the whole purpose of all these time named handling things the JSON JavaScript scripts I realized there are these simple type resolver and so on so those are meant to deal with some of the object object-oriented programming features like polymorphism generics and so on so if you you should be only be sending things like dto some employers write something very simple if you find yourself that you require things like polymorphism then maybe you need to rethink what you're sending Ivano Africa there you go I love this is life sorry just again so yeah this is the the whole design flow in DC analyzation is that they allow you to change the type discriminators and mean that happens then attackers can arbitrary instantiate any type and that's the big problem here any questions alright so thank you [Applause]